

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester VI)
Business Analytics
Practical I(extra)

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Class : TYIT	Date/Time of Submission : 24/07/2026

```
#Import Required Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
#Load dataset
df = pd.read_csv("NetFlix.csv")
```

df.head()

✓ 0.1s Python

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	genres	description
0	s1	TV Show	3%	NaN	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	14-Aug-20	2020	TV-MA	4	International TV Shows, TV Dramas, TV Sci-Fi &...	In a future where the elite inhabit an island ...
1	s10	Movie	1920	Vikram Bhatt	Rajneesh Duggal, Adah Sharma, Indraneil Sengup...	India	15-Dec-17	2008	TV-MA	143	Horror Movies, International Movies, Thrillers	An architect and his wife move into a castle t...
2	s100	Movie	3 Heroines	Iman Brotoseno	Reza Rahadian, Bunga Citra Lestari, Tara Basro...	Indonesia	5-Jan-19	2016	TV-PG	124	Dramas, International Movies, Sports Movies	Three Indonesian women break records by becomi...
3	s1000	Movie	Blue Mountain State: The Rise of Thadland	Lev L. Spiro	Alan Ritchson, Darin Brooks, James Cade, Rob R...	United States	1-Mar-16	2016	R	90	Comedies	New NFL star Thad buys his old teammates' belo...
4	s1001	TV Show	Blue Planet II	NaN	David Attenborough	United Kingdom	3-Dec-18	2017	TV-G	1	British TV Shows, Docuseries, Science & Nature TV	This sequel to the award-winning nature series...

df.tail()

✓ 0.0s Python

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	genres	description
7782	s995	TV Show	Blown Away	NaN	NaN	Canada	12-Jul-19	2019	TV-14	1	International TV Shows, Reality TV	Ten master artists turn up the heat in glassbl...
7783	s996	TV Show	Blue Exorcist	NaN	Nobuhiko Okamoto, Jun Fukuyama, Kana Hanazawa...	Japan	1-Sep-20	2017	TV-MA	2	Anime Series, International TV Shows	Determined to throw off the curse of being Sat...
7784	s997	Movie	Blue Is the Warmest Color	Abdellatif Kechiche	Léa Seydoux, Adèle Exarchopoulos, Salim Kechio...	France, Belgium, Spain	26-Aug-16	2013	NC-17	180	Dramas, Independent Movies, International Movies	Determined to fall in love, 15-year-old Adele ...
7785	s998	Movie	Blue Jasmine	Woody Allen	Cate Blanchett, Sally Hawkins, Alec Baldwin, L...	United States	8-Mar-19	2013	PG-13	98	Comedies, Dramas, Independent Movies	The high life leads to high anxiety for a fash...
7786	s999	Movie	Blue Jay	Alex Lehmann	Sarah Paulson, Mark Duplass, Clu Gulager	United States	6-Dec-16	2016	TV-MA	81	Dramas, Independent Movies, Romantic Movies	Two former high school sweethearts unexpectedl...

df.shape

✓ 0.0s

(7787, 12)

```
#Display Column Names
df.columns
```

✓ 0.0s

```
Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
      'release_year', 'rating', 'duration', 'genres', 'description'],
      dtype='str')
```

```
#Display Dataset Information
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 7787 entries, 0 to 7786
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   show_id         7787 non-null   str
1   type            7787 non-null   str
2   title           7787 non-null   str
3   director        5398 non-null   str
4   cast            7069 non-null   str
5   country         7280 non-null   str
6   date_added      7777 non-null   str
7   release_year    7787 non-null   int64
8   rating          7780 non-null   str
9   duration        7787 non-null   int64
10  genres          7787 non-null   str
11  description     7787 non-null   str
dtypes: int64(2), str(10)
memory usage: 730.2 KB
```

```
#Display Data Types
df.dtypes
```

✓ 0.0s

```
show_id      str
type         str
title        str
director     str
cast         str
country      str
date_added   str
release_year int64
rating       str
duration     int64
genres       str
description  str
dtype: object
```

```
#Check Missing Values
df.isnull().sum()
```

```
show_id      0  
type         0  
title        0  
director    2389  
cast        718  
country     507  
date_added   10  
release_year 0  
rating       7  
duration     0  
genres       0  
description  0  
dtype: int64
```

```
#Check Duplicate Records  
df.duplicated().sum()
```

```
#Display Descriptive Statistics  
df.describe()
```

✓ 0.0s

	release_year	duration
count	7787.000000	7787.000000
mean	2013.932580	69.122769
std	8.757395	50.950743
min	1925.000000	1.000000
25%	2013.000000	2.000000
50%	2017.000000	88.000000
75%	2018.000000	106.000000
max	2021.000000	312.000000

```
#Count Unique Values  
df.nunique()
```

✓ 0.0s

show_id	7787
type	2
title	7787
director	4049
cast	6831
country	681
date_added	1565
release_year	73
rating	14
duration	206
genres	492
description	7769

dtype: int64

```
#Display Unique Values of a Particular Column  
df["cast"].unique()
```

✓ 0.0s

```
<StringArray>  
[
```

```
#Count Product Categories  
df["cast"].value_counts()
```

```
cast  
David Attenborough  
Samuel West  
Jeff Dunham  
Kevin Hart  
Craig Sechler
```

```
Johnny Depp, Penélope Cruz, Franka Potente, Rachel Griffiths, Paul Reubens, Jordi  
Nobuhiko Okamoto, Jun Fukuyama, Kana Hanazawa, Kazuya Nakai, Koji Yusa, Yuki Kaji,  
Léa Seydoux, Adèle Exarchopoulos, Salim Kechiouche, Aurélien Recoing, Catherine Sa  
Cate Blanchett, Sally Hawkins, Alec Baldwin, Louis C.K., Bobby Cannavale, Andrew D  
Sarah Paulson, Mark Duplass, Clu Gulager  
Name: count, Length: 6831, dtype: int64
```

```
#Count Store Locations  
df["country"].value_counts()
```

✓ 0.0s

```
country  
United States                2555  
India                        923  
United Kingdom              397  
Japan                       226  
South Korea                 183  
...  
United Kingdom, Russia, United States    1  
United States, France, Serbia            1  
Germany, Jordan, Netherlands            1  
Argentina, United States                1  
France, Belgium, Spain                  1  
Name: count, Length: 681, dtype: int64
```

```
#Count Holiday Effect
df["cast"].value_counts()
```

✓ 0.0s

cast	# count
David Attenborough	18
Samuel West	10
Jeff Dunham	7
Kevin Hart	6
Michela Luci, Jamie Watson, Eric Peter	6
Craig Sechler	6
Bill Burr	5
Iliza Shlesinger	5
Jim Gaffigan	5
Mike Birbiglia	4

```
df["release_year"].mean()
```

✓ 0.0s

np.float64(2013.932579940927)

```
df["release_year"].max()
```

✓ 0.0s

np.int64(2021)

```
df["release_year"].min()
```

✓ 0.0s

np.int64(1925)

```
df.corr(numeric_only=True)
```

✓ 0.0s

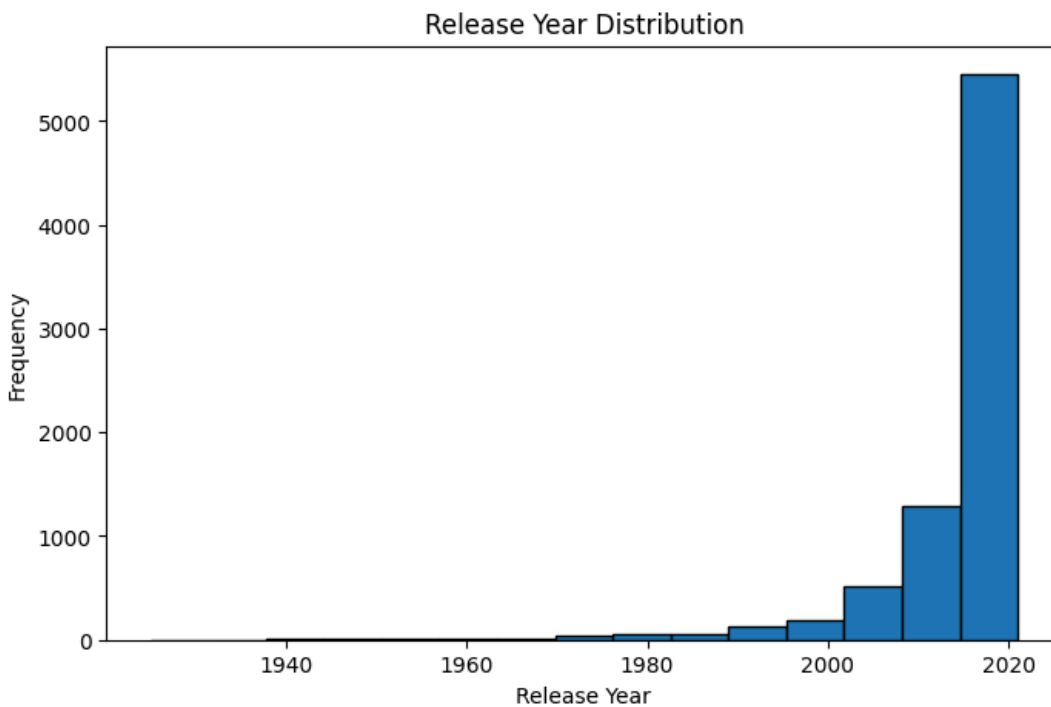
	# release_year	# duration
release_year	1.0	-0.24083598848535492
duration	-0.24083598848535492	1.0

```
numeric_df = df.select_dtypes(include=["number"])
numeric_df.corr()
```

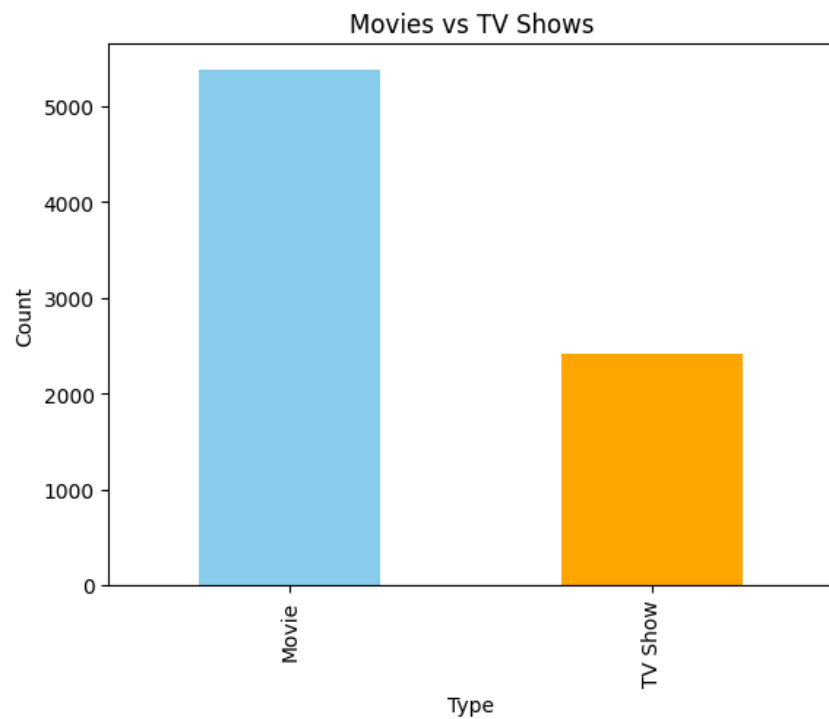
✓ 0.0s

	# release_year	# duration
release_year	1.0	-0.24083598848535492
duration	-0.24083598848535492	1.0

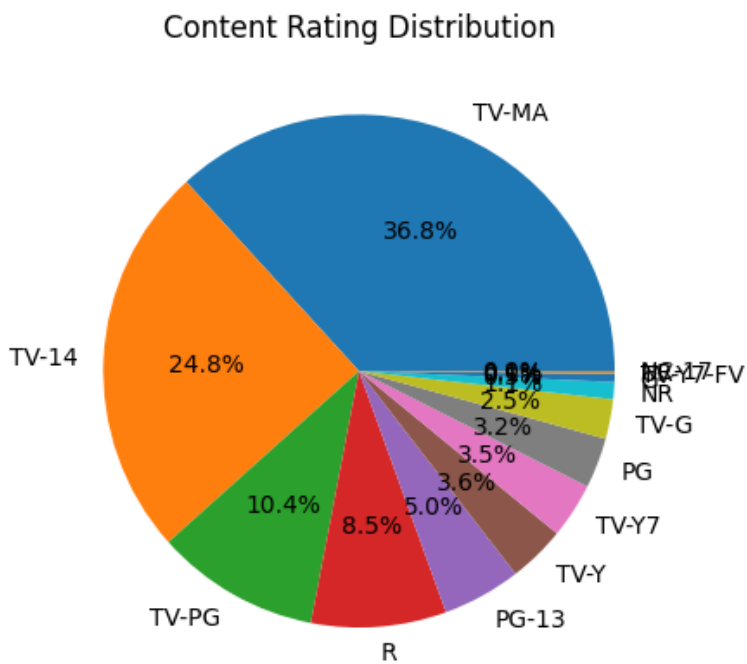
```
plt.figure(figsize=(8,5))
plt.hist(df["release_year"], bins=15, edgecolor="black")
plt.title("Release Year Distribution")
plt.xlabel("Release Year")
plt.ylabel("Frequency")
plt.show()
```



```
df["type"].value_counts().plot(kind="bar", color=["skyblue", "orange"])
plt.title("Movies vs TV Shows")
plt.xlabel("Type")
plt.ylabel("Count")
plt.show()
```

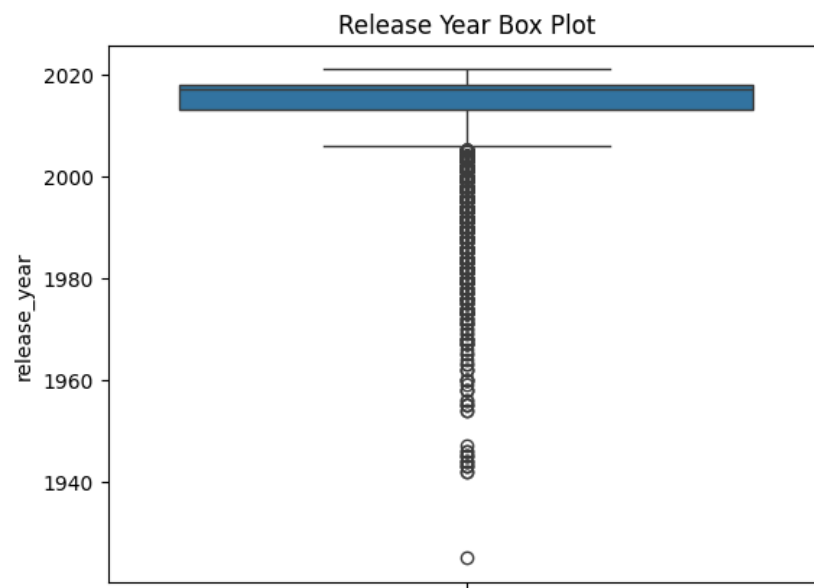


```
df["rating"].value_counts().plot(kind="pie", autopct="%1.1f%%")  
plt.title("Content Rating Distribution")  
plt.ylabel("")  
plt.show()
```



```
sns.boxplot(y=df["release_year"])  
plt.title("Release Year Box Plot")
```

```
plt.show()
```



```
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="Blues")  
plt.title("Correlation Heatmap")  
plt.show()
```

